CH-24 Ergonomics Homework Answers

01

What is the lifting index for an operator to lift a 12kg weight and a recommended weight limit is 12.1 kg?

A. 0.526

B. **0.992**

C. 1.008

D. 1.901

Lifting Index = Weight to be lifted / Recommended Weight

LI = 12kg / 12.1kg

02

When using the NIOSH lifting equation, you will be able to determine the recommended weight limit for lifting tasks using the formula developed through research in ergonomics. If the actual load is observed and divided by the recommended weight limit, you will calculate:

A. The optimum task multiplier

B. The severity rate for the task

C. The percent of the population that can do the task

D. The lifting index

03

When attempting to reduce ergonomic injuries, which of the following actions is the most proactive?

A. **Design reviews**

B. Injury/illness data reviews

C. Medical surveillance

D. Audits/assessments

04

All of the following are administrative controls that could be considered for the reduction or prevention of ergonomic injuries EXCEPT:

A. Job expansion

B. Tool changes

C. Job rotation

D. Rest periods

05

Which work scenario would benefit the most from an ergonomic study?

A. Using self-retracting lanyards for fall protection.

B. Operating an electric drill to install a door jamb.

C. Lifting materials from floor level to overhead for installation.

D. Wearing waders while working with concrete.

06

Tools are well-designed when they:

A. Fit within statistical reliability parameters

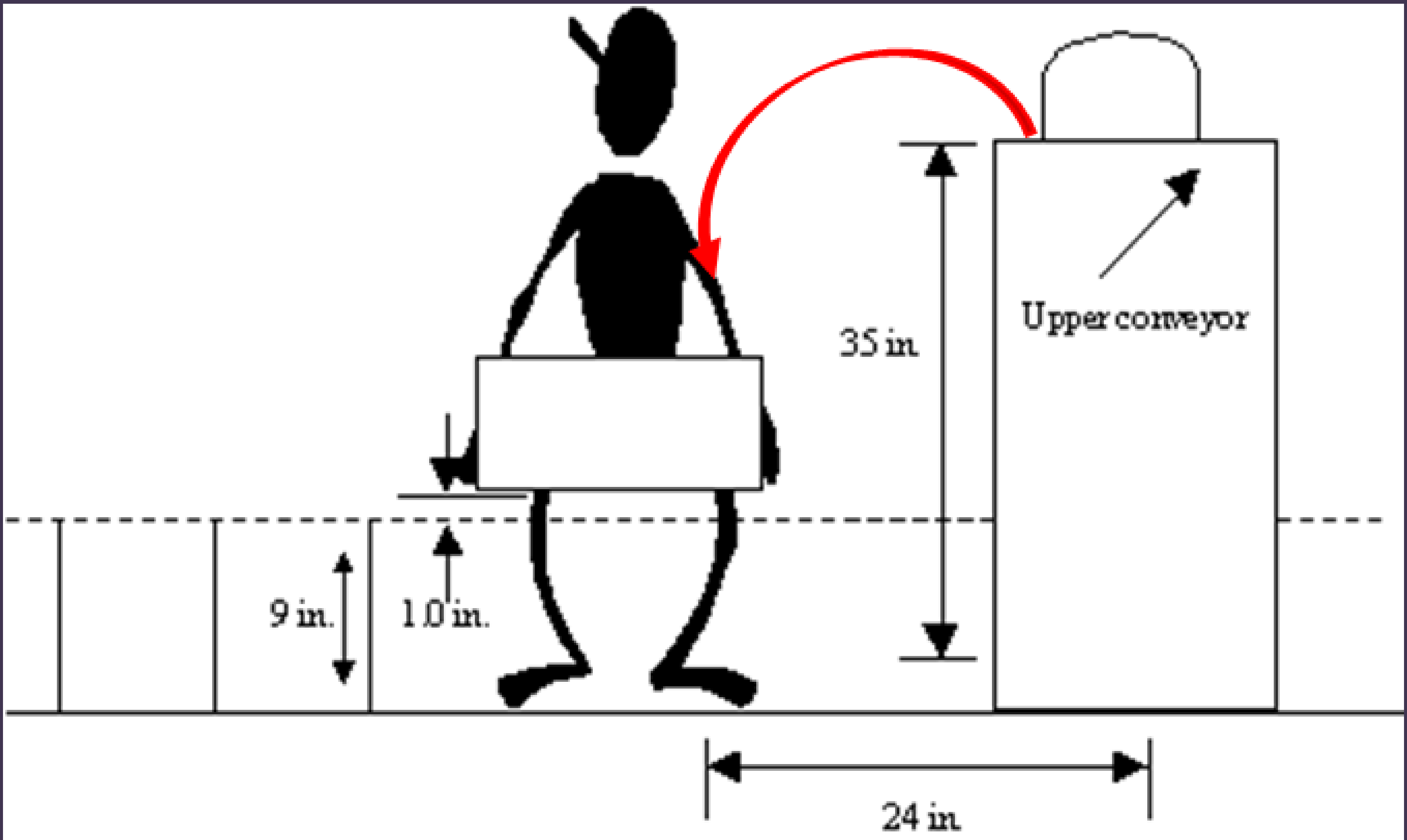
B. Allow for neutral postures

C. Require static work

D. Have handles larger than the worker's hands

07

You are evaluating a lifting task in your facility that involves moving tote boxes containing parts from an upper conveyor to a lower conveyor. The totes are poorly constructed and are difficult to grasp. Most workers simply grasp the bottom of the slippery tote box from the upper conveyor (located 35 in. from the floor) and drop the tote box 1 in. from the lower conveyor (located 9 in. from the floor). The conveyors are at right angles to each other, and the centre of the conveyor belts are 24 in. from the feet of the workers. Two lifts are required every minute during a 2-hr shift. The tote box weighs 4 lb. If the weight of parts placed in the tote box is 8 lb, what is the lifting index? Should changes be made?


Upper conveyor
35 in.
9 in.
10 in.
24 in.

$RWL = LC * HM * VM * DM * AM * FM * CM$

LC = 51 lb, HM = (10/H), VM = [1-0.0075 * absolute value of (V-30)]

DM = [0.82 + (1.8/D)], AM = (1-0.0032 * A)

These are the numbers we will use in the RWL equation.

H = 24 in. . . . this is the horizontal distance from the feet

V = 35 in. . . . this is the vertical distance from the floor

D = 25 in. . . . this is the lifted distance (the workers drop the tote the last inch - that is why D is not 26 in.)

A = 90º . . . this is the asymmetry (body twisting)

FM = 0.84 ... this is the frequency factor

CM = 0.90 ... this is the coupling factor

Frequency Multiplier

Frequency, Lifts/min (*F*)[a]	Work Duration					
	≤1 h		>1 h but ≤2 h		>2 h but <8 h	
	$V < 30$[b]	$V \geq 30$	$V < 30$	$V \geq 30$	$V < 30$	$V \geq 30$
≤ 0.2	1.00	1.00	0.95	0.95	0.85	0.85
0.5	0.97	0.97	0.92	0.92	0.81	0.81
1	0.94	0.94	0.88	0.88	0.75	0.75
2	0.91	0.91	0.84	$\boxed{0.84}$	0.65	0.65
3	0.88	0.88	0.79	0.79	0.55	0.55
4	0.84	0.84	0.72	0.72	0.45	0.45
5	0.80	0.80	0.60	0.60	0.35	0.35

Coupling Multiplier

Coupling Type	Coupling Multiplier	
	$V < 30$ in. or 75 cm	$V \geq 30$ in. or 75 cm
Good	1.00	1.00
Fair	0.95	1.00
Poor	0.90	$\boxed{0.90}$

RWL = LC * HM * VM * DM * AM * FM * CM

RWL = 51 lb (10/24) [1 - 0.0075 x (absolute 35 -30)] [0.82 + (1.8/25)] [1 -(0.0032 * 90)] * 0.84 * 0.90

= 51 lb * 0.417 * (0.9625) * 0.892 * 0.712 * 0.84 * 0.90 = **9.8 lb**

Total load = 4.0 lb + 8.0 lb = **12 lb**

Lifting Index = Load / Recommended Weight Limit

= 12 lb / 9.8 lb = **1.2**

A lifting index of over 1.00 is too high and indicates that changes to the process and/or to the magnitude of the load should be made to reduce the lifting index to 1.00 or less.

07

A. 1.22/Changes should be made

B. 0.81/Changes are not necessary

C. 0.78/Changes are not necessary

D. 1.15/Changes should be made